

Smart Public Transport Analytics Data Dictionary

Teyzix Core - Data Analytics Internship - Task DA-3

1. Dataset Overview

The following table contains the information about the dataset.

Attribute	Detail
Dataset name	transport_cleaned.csv
Number of records	1,300
Number of fields	15
Time period	January – April 2025 (4 months)
Routes covered	8 (R01–R08)
Drivers	40 (plus “Unknown” for unrecorded)
Source	Self-generated (synthetic) data simulating transport operations
Currency	Pakistani Rupee (Rs)

The dataset captures four months of simulated daily operations for a public transport network. Each of the 1,300 records represents a single bus trip, linking a route and vehicle to its schedule, passenger load, revenue, fuel use, and operational outcome. Together the 15 fields describe both the *service side* of each trip (route, bus, timing, delay, status) and its *business side* (passengers, revenue, fuel), allowing analysis of efficiency, demand, and profitability from a single, consistent source.

2. Field Definitions

The dataset contains 15 fields. Each is described below with its data type, meaning, and permitted values or range.

Field Name	Data Type	Description	Allowed Values / Range
Route ID	Text (categorical)	Unique identifier for the transport route.	R01, R02, R03, R04, R05, R06, R07, R08
Route Name	Text	Descriptive name of the route (origin – destination).	e.g. City Center – Airport
Bus Number	Text	Identifier of the individual bus vehicle used.	Format: BUS-NNN (e.g. BUS-309)
Bus Type	Text (categorical)	Category of the bus, determining seating capacity.	Standard (50), AC (45), Electric (40), Mini (25)
Date	Date	Calendar date of the trip.	YYYY-MM-DD (2025-01-01 to 2025-04-30)
Departure Time	Time	Scheduled departure time of the trip.	24-hour HH:MM (00:00–23:59)
Arrival Time	Time	Actual arrival time of the trip.	24-hour HH:MM
Delay Duration	Integer	Minutes the trip arrived later than scheduled.	0 – 63 minutes
Passenger Count	Integer	Number of passengers carried on the trip.	10 – 50 (never exceeds bus capacity)
Ticket Revenue	Decimal	Total fare revenue collected for the trip (Rs).	600.00 – 6,864.00
Fuel Consumption	Decimal	Fuel used for the trip, in litres.	0.00 – 18.76 (Electric buses ≈ 0)
Driver ID	Text (categorical)	Identifier of the driver assigned to the trip.	D001 – D040, or “Unknown”
Weather Condition	Text (categorical)	Weather during the trip.	Clear, Cloudy, Fog, Rain, Storm
Day Type	Text (categorical)	Whether the trip occurred on a weekday or weekend.	Weekday, Weekend
Trip Status	Text (categorical)	Final operational outcome of the trip.	Completed, Delayed, Cancelled

3. Notes

Bus capacity: Passenger Count is bounded by the capacity of the Bus Type (Standard 50, AC 45, Electric 40, Mini 25); no trip exceeds its vehicle’s capacity.

Derived relationship: Ticket Revenue is a function of Passenger Count and the route’s fare; Arrival Time equals Departure Time plus scheduled duration plus Delay Duration.

“Unknown” values: the Driver ID field may contain “Unknown”, representing records where the driver was not recorded and the missing value was retained explicitly during cleaning rather than dropped.

Data origin: this is a self-created (synthetic) dataset generated to simulate realistic transport operations for the project. It contains no real personal or operational data.